THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · EASY LEVEL UNIT TEST SET 1

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The complement of 30° is: 1

(a) 150° **(b)** 30° **(c)** 120° **(d)** 60°

2. The sum of two odd numbers is always: 1

(a) even **(b)** odd **(c)** a prime number **(d)** a multiple of 5

3. Can a triangle have sides 3 cm, 4 cm and 5 cm? 1

(a) No **(b)** Yes

(c) Only if it is isosceles **(d)** Only if it is right-angled

4. The product of $\frac{2}{3}$ and $\frac{3}{5}$ is: 1

(a) $\frac{1}{5}$ **(b)** $\frac{4}{5}$ **(c)** $\frac{5}{8}$ **(d)** $\frac{2}{5}$

5. The supplement of 50° is: 1

(a) 40° **(b)** 130° **(c)** 310° **(d)** 50°

6. The sum of the digits of 47 is: 1

(a) 10 **(b)** 12 **(c)** 11 **(d)** 13

7. Two angles of a triangle are 50° and 60°. The third angle is: 1

(a) 70° **(b)** 80° **(c)** 60° **(d)** 90°

8. $\frac{2}{5}$ of 20 is: 1

(a) 50 **(b)** 8 **(c)** 4 **(d)** 16

9. Two lines intersect and one of the angles formed is 40°. The angle vertically opposite to it is: 1

(a) 50° **(b)** 40° **(c)** 60° **(d)** 30°

10. The reverse of 3475 is: 1

(a) 4753 **(b)** 3457 **(c)** 5743 **(d)** 3475

11. A triangle with all three sides equal is called: 1

(a) isosceles **(b)** right-angled **(c)** scalene **(d)** equilateral

12. The reciprocal of $\frac{3}{7}$ is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{3}{7}$ **(c)** $-\frac{7}{3}$ **(d)** $\frac{7}{3}$

13. Two angles form a linear pair and one of them is 60°. The other is: 1

(a) 130° **(b)** 140° **(c)** 110° **(d)** 120°

14. Which of these numbers is a palindrome? 1

(a) 1313 **(b)** 1231 **(c)** 1331 **(d)** 1332

15. In a triangle, the two interior opposite angles are 40° and 55°. The exterior angle is: 1

(a) 95° **(b)** 115° **(c)** 105° **(d)** 85°

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the complement of 35° and the supplement of 110°. 2

17. Write the Collatz sequence that starts from 6 (halve if even; triple and add 1 if odd) until you reach 1. 2

18. Can a triangle be formed with sides 6 cm, 8 cm and 10 cm? Give a reason. 2

19. Multiply $\frac{3}{5}$ by $\frac{10}{21}$ and write in the lowest terms. 2

20. Two lines intersect and one of the four angles is 65°. Find the other three angles. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Two angles of a linear pair are in the ratio $1 : 2$. Find them. 3

22. Apply one step of Kaprekar's routine to 3524 (largest arrangement of the digits minus smallest arrangement). 3

23. The angles of a triangle are in the ratio $1 : 2 : 3$. Find the angles. 3

24. Find $2\frac{1}{3} \times \frac{3}{5}$. 3

OR

The angles x° and 110° are supplementary. Find x and say whether the angle x° is acute, right or obtuse. 3

25. The sum of three consecutive numbers is 48. Find the numbers. 3

OR

The apex angle of an isosceles triangle is 30°. Find the base angles. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Two lines intersect at O and one angle is 35°. Find the other three angles and show that the four angles add up to 360°. 5

OR

Take the two-digit number 26. Add it to its reverse, show that the sum is divisible by 11 and find the quotient. 5

27. A triangle has sides 5 cm, 5 cm and 8 cm. Check that it exists, name it by its sides and find its perimeter. 5

OR

A tank holds 120 litres. It is $\frac{3}{4}$ full. $\frac{2}{3}$ of the water in it is then used. Find the water at first, the water used and the water left. 5

END OF QUESTION PAPER — **Total: 50 Marks**